

(Strictly-)Supermodular Games

(aka *Games with Strategic Complementarities*)

Consider a family of simultaneous-move strategic form games $\{G(t)\}_{t \in T}$, where $\emptyset \neq T \subset \mathbb{R}^\ell$, $\ell \geq 1$, and $G(t) = (N, \{A_i\}_{i \in N}, \{u_i(\cdot, t)\}_{i \in N})$, where for $i \in N$, and $u_i : A \times T \rightarrow \mathbb{R}$ and A is the Cartesian product of the strategy sets A_i . Impose these assumptions hold for $i \in N$:

1. The strategy set A_i is a nonempty, compact subset of $\mathbb{R}$.
2. The function $u_i(\cdot, a_{-i}, t)$ is continuous on A_i for every $(a_{-i}, t) \in A_{-i} \times T$.
3. The function u_i satisfies strictly increasing differences in a_i and (a_{-i}, t) .¹

Call a family of games satisfying 1-3 *strictly-supermodular (SSPM) games*.² Assumptions 1 and 2 ensure that the set of player i 's best replies is non-empty and compact for every $(a_{-i}, t) \in A_{-i} \times T$ and Assumptions 1-3 together imply that there is a *largest* and *smallest* best reply for every $(a_{-i}, t) \in A_{-i} \times T$. (Why?) Let $r_i(\cdot, t)$ be player i 's best reply correspondence, let $\bar{r}_i(\cdot, t)$ give player i 's *largest* best reply and $\underline{r}_i(\cdot, t)$ give player i 's *smallest* best reply in $G(t)$. It follows from Topkis's Theorem in the *Monotone Comparative Statics* notes that both $\bar{r}_i(\cdot, \cdot)$ and $\underline{r}_i(\cdot, \cdot)$ are increasing functions.

The main tool is *Tarski's fixed point theorem* on the existence of largest and smallest fixed points.

Lemma. (Corollary to Tarski's Fixed Point Theorem)³ Let $X \neq \emptyset$ be a compact sublattice of $\mathbb{R}^m$, and suppose that $f : X \rightarrow X$ is an increasing function: for x'' and x' in X with $x'' \geq x'$, $f(x'') \geq f(x')$. Then f has a smallest x_* and a largest x^* fixed point: that is, x_* and x^* are fixed points of f , and for any fixed point x of f , $x_* \leq x \leq x^*$. Moreover, the largest fixed point is $x^* = \sup\{x \in X \mid f(x) \geq x\}$ and the smallest is $x_* = \inf\{x \in X \mid f(x) \leq x\}$.

Proof of the Lemma: Let $\bar{X} = \{x \in X \mid f(x) \geq x\}$ and let x^* be the pointwise supremum of $\bar{X}$.⁴

$$x^* = \sup\{x \in X \mid f(x) \geq x\}. \quad (1)$$

Since X is a nonempty, compact sublattice of a Euclidean space, X has an infimum, $\inf X$, and $\inf X \in X$, so $f(\inf X) \geq \inf X$, and $\bar{X}$ is nonempty; moreover $x^* \in X$ (why?). For any $x \in \bar{X}$

$$x \leq f(x) \leq f(x^*) \quad (2)$$

since $x \leq x^*$ and f is increasing. From (1) and (2), $x^* \in \bar{X}$, so $f(x^*) \geq x^*$. Since $f(x^*) \geq x^*$ and f is increasing, $f(f(x^*)) \geq f(x^*)$ and so $f(x^*) \in \bar{X}$. But then (1) implies that $f(x^*) \leq x^*$. So $f(x^*) = x^*$. Since $\bar{X}$ contains all the fixed points of f , (1) implies that x^* is the largest fixed point of f . The argument for the smallest fixed point is similar. $\square$

¹A *sufficient* condition for 3. is that u_i satisfies *pairwise* strictly increasing differences for every pair of real variables that are arguments of u_i .

²See Section 10.3 of Sundaram's text, Milgrom and Roberts (1990) and Chapter 2 in Vives (1999). Milgrom and Shannon (1994) extend the analysis to the case of games satisfying the SSP. (Full citations are in the reading list in the Syllabus.)

³Tarski, A. "A Lattice-Theoretical Fixpoint Theorem and Its Applications." *Pacific J. Math.* 5, 285-309, 1955, Theorem 1. Here I just copy out his elegant, concise proof, specialized to the case of Euclidean X .

⁴How do we know that the supremum exists? Let $\bar{x}_i = \sup\{x_i \mid x \in X\}$, which certainly exists since $\{x_i \mid x \in X\}$ is compact. For each $1 \leq i \leq m$, there is a sequence $x_i^{j_n}$ with $\lim_{n \rightarrow \infty} x_i^{j_n} = \bar{x}_i$. Let $x^n = x_1^{j_n} \vee x_2^{j_n} \vee \dots \vee x_m^{j_n}$. Then for $i = 1, \dots, m$, $\bar{x}_i \geq x_i^n \geq x_i^{j_n}$. Since $x_i^{j_n} \rightarrow \bar{x}_i$, it follows that $\lim_{n \rightarrow \infty} x^n = \bar{x} = (\bar{x}_1, \dots, \bar{x}_m)$. Since X is compact, $\bar{x} \in X$ and $\bar{x} \geq x$ for every $x \in X$. A similar argument shows that X has a pointwise infimum in X .

Proof of the lemma for $X \subset \mathbb{R}$. Let $\overline{X} = \{x \in X \mid f(x) \geq x\}$. Since X is a nonempty, compact subset of $\mathbb{R}$, X has an infimum, $\underline{x}$, and $\underline{x} \in X$, so $f(\underline{x}) \geq \underline{x}$, implying $\overline{X}$ is nonempty. Let x^* be the supremum of $\overline{X}$:

$$x^* = \sup\{x \in X \mid f(x) \geq x\}. \quad (3)$$

Since X is compact, the supremum exists in X . For any $x \in \overline{X}$, we have

$$f(x^*) \geq f(x) \geq x,$$

where the first inequality follows from f increasing, so $f(x^*) \geq x$ for every $x \in \overline{X}$. Take limits as $x \rightarrow x^*$ for any sequence of points in $\overline{X}$ that converge to x^* to find

$$f(x^*) \geq x^*. \quad (4)$$

Since f is increasing and $f(x^*) \in X$, it follows from (4) that $f(f(x^*)) \geq f(x^*)$, and so $f(x^*) \in \overline{X}$. (Whoa!) But then (3) implies that $f(x^*) \leq x^*$. This inequality and (4) imply $f(x^*) = x^*$ so that x^* is a fixed point of f . Since $\overline{X}$ contains all the fixed points of f , (3) implies that x^* is the *largest* fixed point of f . The argument for the smallest fixed point is similar. $\square$

Theorem. (*Existence of Largest and Smallest Nash Equilibria (Topkis, 1979)*) *If assumptions 1-3 hold, then $G(t)$ has a largest a^* and smallest a_* Nash Equilibrium. Moreover $a^*(t) = \vee\{a \in A \mid \bar{r}(a) \geq a\}$ and $a_*(t) = \wedge\{a \in A \mid \bar{r}(a) \leq a\}$.⁵*

Proof of the Existence Theorem: Suppress t as an argument of the best replies and let $\bar{r}(a) = (\bar{r}_1(a_{-1}), \dots, \bar{r}_n(a_{-n}))$ for every $a \in A$. By Topkis's Theorem applied to the *decision problems* which determine best replies, the function $\bar{r} : A \rightarrow A$ is increasing. Since A is a nonempty, compact sublattice of a Euclidean space, the Lemma implies that $\bar{r}$ has a largest fixed point, $a^* \in A$. Clearly $a^* \in NE(G)$. Let $\tilde{a}$ be any other Nash equilibrium of G , so $\tilde{a} \in r(\tilde{a})$. Since the $\bar{r}_i$'s are the *largest* best responses, $\tilde{a} \leq \bar{r}(\tilde{a})$, so $\tilde{a} \in \overline{A} = \{a \in A \mid \bar{r}(a) \geq a\}$. And since a^* is the pointwise supremum of $\overline{A}$, it follows that $a^* \geq \tilde{a}$, and a^* is the largest Nash equilibrium of G . The argument for the existence of a smallest Nash equilibrium is similar. $\square$

If in addition to Assumptions 1-3, the best replies are single-valued, then it follows almost immediately that $G(t)$ has both a largest and a smallest Nash equilibrium: just apply the Lemma to the increasing *function* (suppressing t) $r(a) = (r_1(a_{-1}), \dots, r_n(a_{-n}))$ on the set A of products of compact sets of real numbers.

Compared with the other “classical” Debreu-Fan-Glicksberg (D-F-G) and Nash (1950) existence theorems this theorem does not require the strategy sets to be convex or the utilities to be continuous in the other player's strategies. It applies for example to finite games. Unlike these classical existence theorems, however, it imposes pairwise SID.

Theorem. (*Monotone Comparative Statics in Games*) *If assumptions 1-3 hold, both the smallest Nash equilibrium $a_*(\cdot)$ and largest Nash equilibrium $a^*(\cdot)$ are increasing on T .⁶*

Proof of the Comparative Statics Theorem: Let t', t'' be points in T with $t'' > t'$. We have $a^*(t') = \bar{r}(a^*(t'), t')$ and $\bar{r}(a^*(t'), t') \leq \bar{r}(a^*(t'), t'')$, where the first follows the definition of a^* and the second follows from Topkis's Theorem applied to the decision problems that determine the best replies. But since $a^*(t') \leq \bar{r}(a^*(t'), t'')$ and $a^*(t'') = \vee\{a \in A \mid a \leq \bar{r}(a, t'')\}$, it follows that $a^*(t'') \geq a^*(t')$. So $a^*(\cdot)$ is increasing on T . The argument that $a_*(\cdot)$ is increasing is similar. $\square$

Remark. *The proof of the comparative statics theorem appeals (indirectly) to the compactness of the strategy sets. In some unbounded games this is innocuous; without loss of generality, the unbounded strategy set can*

⁵Topkis, Donald M. “Equilibrium points in nonzero-sum n-person submodular games.” *SIAM Journal on Control and Optimization* 17 (1979): 773-787.

⁶See Theorem 6 in Milgrom and Roberts (1990). They also have several examples.

be replaced with a compact set. In other games, it is not. The conclusion of the comparative statics theorem remains true under weaker conditions than compactness. For example if smallest and largest Nash equilibria **exist** in the game, then the conclusion of the comparative statics theorem holds.⁷

Exercises 2 and 3 will appear on the final ASMT.

Exercise 1. Specialize to the case of $n = 2$ and games $G(t)$ with **single-valued** best replies for $t = 0, 1$. Apply the Lemma to the **function** $f(a_1, t) = r_1(r_2(a_1, t), t)$ from A_1 into A_1 for each fixed $t \in T$ to verify the conclusions of the Existence and Comparative Statics Theorem for this case.

Exercise 2. Consider a **differentiated-products** Bertrand oligopoly game with C^2 demands $d_i(p_1, p_2, \dots, p_n)$ for $i = 1, 2, \dots, n$ and constant unit cost of $k_i > 0$ for firm i . Let $k = (k_1, \dots, k_n)$. Prove that if each firm i 's elasticity of demand is strictly increasing in the price of each firm $j \neq i$, then each firm i 's lowest best reply, $\underline{r}_i(p_{-i}, k)$ is increasing in (p_{-i}, k) in the sense that if $p''_{-i} > p'_{-i}$ and $k'' >> k'$, then $\underline{r}_i(p''_{-i}, k'') \geq \underline{r}_i(p'_{-i}, k')$. Suppose further a Nash equilibrium $\tilde{p}^t$ exists for k^t , $t = 0, 1$, $k^1 >> k^0$. Argue that there exists a smallest NE $\underline{p}^t$ of $G(t)$ and that $\underline{p}^1 \geq \underline{p}^0$.

Exercise 3. Consider a homogenous-products Cournot **duopoly** game with C^2 inverse demand $p(\cdot)$ and cost function $c_i(q_i) = k_i q_i$ for firm i . (a) Are there any difficulties in asserting that each firm i 's profit satisfies strictly increasing differences in (q_i, q_j) ? If not, explain when SID holds, if so, can you reformulate the game so it can hold? (Hint: think change of variables, replacing q_2 with something else.) (b) Show how you can apply the Existence Theorem from the SPM Games Notes to show the existence of largest and smallest Nash equilibria for your reformulated game. (c) For whatever formulation you devise, how does an increase in firm 1's marginal cost affect the largest and smallest Nash equilibrium? (d) Sketch how you might extend the argument beyond the constant-returns case. (e) Are there any difficulties in extending the argument in parts (a) and (b) to three or more firms? If so what are they?

⁷Suppose that largest NE exist at t' and t'' . Consider $A(t', t'') = \{a^*(t'), a^*(t''), a^*(t') \vee a^*(t''), a^*(t') \wedge a^*(t'')\} \subset A$. Clearly $A(t', t'')$ is a compact lattice. Restrict player i 's strategy set to be the i -th component of each of the four vectors in $A(t', t'')$. Now apply the argument of the Existence and Comparative Statics Theorems to the game with these restricted strategy sets.